CHEMISTRY Department
SIMHADRI

Note No :1

Ref:SMTPP/Chemistry/2025-26/ofc note/424772

Date: 18/09/2025(17:25:31)

Subject:LMI-MANAGEMENT OF CONDENSER TUBE LEAKAGE IN SEA WATER COOLED CONDENSER

LOCATION MANAGEMENT INSTRUCTION ON- MANAGEMENT OF CONDENSER TUBE LEAKAGE

IN SEA WATER COOLED CONDENSER :

Revision of LMI LMI/OGN/OPS/CHEM/016 MANAGEMENT OF CONDENSER TUBE LEAKAGE

IN SEA WATER COOLED CONDENSER :

The subject LMI is reviewed and made below mentioned changes as per latest OGN.

Following changes have been done from previous version as described below:

1. Offline monitoring parameters are presented in a table.
2. Recommended laboratory instruments for the offline analysis of low-level cations and anions in steam-water samples are also specified.
3. Action levels have been revised in accordance with **COS-OD-CHEM-001** guidelines.
4. Immediate shutdown conditions have been updated in accordance with **COS-OD-CHEM-001** guidelines.
5. Updated the activities to be conducted under Action Levels 1 and 2.
6. Updated Clause 9.0 -Records to be maintained.
7. Updated Check List I and II to capture all relevant information during cycle chemistry disturbances. Additionally, both reports are to be signed by the respective departments as specified in the above.

Encl:

1. LMI in word and .PDF Formats.

2. Latest OGN-OPS-CHEM-016_Rev4_Guidelines on Action to be taken in the event of condenser tube

Leakage.

Submitted for approval please.

Submitted by:

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Note No:2

The draft LMI is submitted for comments from related departments and for final approval please.

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Note No:3

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Note No:4

LMI is in line with the revised OGN. Forwarded for further review.

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Note No:5

Please review the LMI

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Note No:6

LMI reviewed and is in line with the revised OGN.

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Note No:10

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Note No:11

LMI is in line with the OGN. May please be approved.

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Unauthorized Use Prohibited

Note No:13

May please approve.

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Note No:14

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NTPC SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

MANAGEMENT OF CONDENSER TUBE LEAKAGE IN SEA WATER COOLED CONDENSER

LMI/OGN/OPS/CHEM/016
Rev. No.: 08

Approved for Implementation by: HOP (SIMHADRI)

TITLE	MANAGEMENT OF CONDENSER TUBE LEAKAGE IN SEA WATER COOLED CONDENSER
LMI No.	LMI/OGN/OPS/CHEM/016 REV:8

LMI APPROVAL DOCUMENT

1	LMI NAME	MANAGEMENT OF CONDENSER TUBE LEAKAGE IN SEA WATER COOLED CONDENSER.
2	LMI No.	LMI/OGN/OPS/CHEM/016 Rev. No.: 08
3	Reference Document	COS-ISO-00-OGN/OPS/CHEM/16 Rev. No.:4, June. 2025
4	Superseded Document	LMI/OGN/OPS/CHEM/016 Rev. No.: 07
5	Prepared By	V. Lalithadevi Dy.Mgr (Chem)
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		Anil Kumar Maurya AGM (C&I)
7	Reviewed By	Ahmad Kaunain Raza GM O&M
8	Approved By	Sameer Sharma–HOP

TITLE	MANAGEMENT OF CONDENSER TUBE LEAKAGE IN SEA WATER COOLED CONDENSER
LMI No.	LMI/OGN/OPS/CHEM/016 REV:8

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TITLE	MANAGEMENT OF CONDENSER TUBE LEAKAGE IN SEA WATER COOLED CONDENSER
LMI No.	LMI/OGN/OPS/CHEM/016 REV:8

MANAGEMENT OF CONDENSER TUBE LEAKAGE IN SEA WATER COOLED CONDENSER

1.0 INTRODUCTION

Condenser Tube leakage is one of the most serious problems largely upsetting the cycle chemistry. Timely action for cycle chemistry management and attending the leakages becomes necessary to minimize the consequences of ingress of cooling water into the system. Monitoring of online parameters coupled with some offline analyses of chemical parameters are utilized for diagnosis of condenser tube leakage. Boiler water regime requires careful monitoring control and adjustment whenever condenser tube leakage is suspected or confirmed. The condenser tube leakage brings large number of impurities into the condensate and consequently into the water steam circuit. This can be more predominant in sea water cooled power stations where large amounts of chlorides and other salts get into the system during condenser seepage/leakage.

This document shall be of immense use for taking necessary corrective action in case of condenser tube leakage.

2.0 SUPERSEDED DOCUMENTS

Management of condenser tube leakage in sea water cooled condenser LMI/OGN/OPS/CHEM/016 Issue No:1, Revision-7 February 2023

3.0 RECORDS OF REVISION

Following changes have been done from previous version as described below:

a. Offline monitoring parameters are presented in a table.

b. Recommended laboratory instruments for the offline analysis of low-level cations and anions in steam-water samples are also specified.

c. Action levels have been revised in accordance with **COS-OD-CHEM-001** guidelines.

d. Immediate shutdown conditions have been updated in accordance with

COS-OD-CHEM-001 guidelines.

e. Updated the activities to be conducted under Action Levels 1 and 2.

f. Updated Clause 9.0 -Records to be maintained.

g. Updated Check List I and II to capture all relevant information during cycle chemistry disturbances. Additionally, both reports are to be signed by the respective departments as specified in the above.

TITLE	MANAGEMENT OF CONDENSER TUBE LEAKAGE IN SEA WATER COOLED CONDENSER
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4.0 SCOPE

The LMI prescribes the specific instructions to be taken by Operation, maintenance and Chemistry in the event of a condenser tube leakage to eliminate/ reduce problems in pre boiler, boiler and super heater systems.

This document specifies the methods of detection (by online and offline monitoring), interpretation of the analytical values and action to be taken in the event of condenser tube leakage.

5.0 GUIDELINES FOR IDENTIFICATION OF CONDENSER TUBE LEAKAGE (CTL) FOR OPERATING UNITS

5.1 ONLINE MONITORING OF CYCLE CHEMISTRY PARAMETERS IN CASE OF SUSPECTED CTL

Sample	Parameters	Observations
Condensate water	Cation conductivity (CC)/ Degassed cation conductivity (DCC)	The value of CC/ DCC of condensate starts increasing from the recommended values.
	Sodium	Sodium value of condensate shows an increasing trend from the recommended value.
Hotwell (L/R) pass	Specific conductivity	The specific conductivity of both passes, or either one pass (L/R), shows an increasing trend depending on the extent of tube leakage or ingress of contaminant.
Feed water	Cation conductivity (CC)	The CC of the feed water sample shows an increasing trend from the recommended value. For units with a Condensate Polishing Unit (CPU), if the condensate flow through CPU is not maintained at 100%, changes in CC may be observed in the feed water and steam samples
Boiler water	pH	The pH of boiler water starts drifting down* depending on the extent of tube leakage or contaminant ingress.
	Chloride	Chloride value shows an increasing trend from the recommended value
	Silica	Silica value shows an increasing trend from the recommended value

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COMMON NOTE:

A comparison between Condensate CC and DCC values should be conducted to determine the contribution of condenser air in-leakage. This will help rule out condenser seepage or leakage if the DCC values are found to be normal.

5.2 OFFLINE MONITORING OF CYCLE CHEMISTRY PARAMETERS IN CASE OF SUSPECTED CTL

If any of the above parameters (Sec 5.1) show the respective trends as mentioned, the site Chemistry team should monitor the following parameters on an hourly basis to further assess the trend, as long as the parameters remain within the action levels defined in Sec 6.0.

Parameters	Drum boiler units
Silica	Condensate water, CPU service vessels, Feed water and Boiler water
Hardness	Condensate water, Feed water and Boiler water
Chloride	Boiler water (Using Spectro photo meter)
Chloride & Sodium	Condensate water, CPU service vessels, Feed water, Saturated steam and Main steam. (Using Ion chromatograph)

Note:
1.Offline CC of Condensate water, individual CPU service vessels, feed water, and main steam samples should be monitored at regular intervals to validate the online CC analysers.
2.The need for CBD operation to maintain parameters should be closely monitored.
3.The variation in condensate flow with unit load should also be closely monitored.
4.Hardness is generally not detected in the case of minor seepage or sweating in condenser tubes.
Based on the trends of online parameters (both local and DCS values), along with offline test results, the site Chemistry team should confirm the presence of condenser seepage or leakage in the unit under consideration.

6.0 ACTION LEVELS

In the event of condenser tube leakage/seepage or contaminants ingress, the following action levels are defined based on condensate parameters and unit shutdown criteria related to excursions in feed, boiler and steam chemistry parameters.

<table>
<tr><th>Parameter</th><th>Unit type</th><th>Normal level</th><th>Action level 1</th><th>Action level 2</th><th>Action level 3</th><th>Immediate Shutdown</th></tr>
<tr><td colspan="7">Sample name: Condensate water</td></tr>
<tr><td rowspan="2">CC /DCC at 25oC (µS/cm)</td><td>Mixed metallurgy/ WHRB</td><td rowspan="3">Refer COS-OD-CHEM-001</td><td>> 0.3 ≤0.50</td><td>>0.5 ≤0.80</td><td>>0.8 ≤1.0</td><td rowspan="3">If any one of the following conditions becomes true:
1.Boiler water pH (measured at 25o C) falls to 8.0 with a further descending trend.
2.Boiler water pH (measured at 25 C) greater than 11.0.
3.When Main steam CC >1.0µS/cm or Sodium >20ppb</td></tr>
<tr><td>All-ferrous</td><td>>0.2 ≤0.35</td><td>> 0.35 ≤0.5</td><td>> 0.5 ≤1.0</td></tr>
<tr><td>Sodium (ppb)</td><td>Mixed metallurgy/ WHRB/ All-ferrous</td><td>> 2.0 ≤4.0</td><td>> 4.0 ≤8.0</td><td>> 8.0</td></tr>
</table>

6.1 SPECIFIC CRITERIA FOR ACTION LEVELS

An action-level alarm should be initiated based solely on condensate parameters. However, in units equipped with a CPU, the initiation of the appropriate action level should consider both the extent of contaminant ingress and the CPU outlet parameters.

The specific action-level responses for condensate contamination are as follows:

ACTION LEVEL 1: There is a potential for the accumulation of contaminants and corrosion. Return values to normal levels within 48 hours.

i)Flow to the attemperators should be reduced or adjusted sufficiently to maintain sodium levels in steam within the recommended limits. This may require a minor load reduction to avoid overheating.
ii)The boiler water chemistry parameters should be maintained within the recommended limits through controlled HP dosing and boiler blowdown.
ACTION LEVEL 2: A serious disturbance in chemical control has resulted in the accumulation of impurities and corrosion. Corrective action measures are required to return the parameter(s) to Action Level 1 within 12 hours.
The following actions should also be taken to maintain the health of the boiler, turbine, and feedwater heaters:

i Termination of flow to attemperators.
ii Increase the boiler blowdown rate.
iii Increase the condensate polisher regeneration frequency.
iv Plan to address condenser leakage at the first convenient opportunity.

ACTION LEVEL 3: A very serious disturbance requiring substantial intervention, such as load reduction or plant shutdown. This action level indicates that the unit should be prepared for shutdown by reducing load and preparing systems and equipment for shutdown, unless the parameter(s) can be returned to Action Level 2 within 6 hours.
IMPORTANT NOTE:
It should be noted that during load reduction activities with the isolation of the leaking pass, the unit must operate under the condition that 100% of the condensate passes through the CPU to maintain feed water purity within the recommended limits. The CPU bypass valve should always remain in a closed condition, including during the resin transfer period.

7.0 GENERAL RECOMMENDATIONS FOR MONITORING AND ADDRESSING CTL OR INGRESS OF CONTAMINANTS IN THE STEAM-WATER CYCLE

SNo	Recommendations	Responsibility
1	Display of Condensate water CC or DCC meter in unit control room for continuous monitoring and quick decision.	C&I
2	SWAS online analyzers (such as CC, DCC, Sodium, etc.) should always be maintained in proper working conditions to ensure accurate and reliable monitoring.	C&I
3	Availability of standby cation columns filled with fresh cation resin.	Chemistry
4	Uniform measurement range for SWAS online analyzers, as outlined in COS-OIN-SYST-087.	C&I
5	Availability of Hotwell conductivity meters (Left and Right passes) along with PI tags.	C&I
6	Hotwell conductivity analyzers should be calibrated during opportunity or unit overhauls, and pass-wise mapping of the analyzers should also be conducted and verified against their DCS values. A record is to be maintained. Frequency: Unit OH or Once/2-years.	Operation/ Chemistry/ C&I

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7	The makeup lines for the ammonia, tanks should be provided from the DM makeup water header.	TMD
8	The availability of DM water filling to the deaerator from boiler fill pump should be ensured as per the original design.	Operation/ TMD
9	The availability of ammonia dosing lines to deaerator filling line (from boiler fill pump) should be ensured to maintain recommended FST water pH whenever DM makeup is directly added to the deaerator for flushing purposes.	TMD
10	Condenser CW pass isolating valves should be kept in good operating conditions to ensure they can be closed quickly and fully without any issues.	Operation/ TMD
11	The FST drain water should not be collected or diverted into the condenser minus level pit, as this is required to ensure the draining of both the FST and the Hotwell simultaneously.	Operation/ TMD
12	In view of the highly corrosive nature of seawater, it is crucial to have clear and specific instructions for steam-water cycle chemistry management. This ensures the prevention of corrosion and damage to the boiler and feed water cycle system of the unit in the event of a condenser tube leakage.	Operation
13	For seawater-cooled condenser units, the action level alarm for condenser tube leakage should be triggered based solely on condensate parameters. However, the appropriate action level may be determined based on the extent of tube leakage and CPU outlet parameters.	Operation/ Chemistry/ C&I
14	To ensure the timely stopping of units in the event of condenser tube leakage (to prevent seawater ingress into the feed system and boiler), all coastal-based plants should be equipped with an independent alarm with a hooter facility, which will activate if any three out of the following five conditions become true: i. Condensate CC > 3 μS/cm ii. CPU inlet CC > 3 μS/cm iii. CPU-A outlet CC > 3 μS/cm iv. CPU-B outlet CC > 3 μS/cm v. Eco inlet CC > 3 μS/cm Note: During CPU changeover, the above logic is to be bypassed and records to be maintained for the same on all occasions.	Operation/ C&I
15	Cycle water chemistry parameters to be maintained within specified limits to prevent damage to boiler and steam cycle components of the unit whenever condenser tube leakage or ingress of contaminants is suspected.	Operation/ Chemistry
16	Whenever a unit is coming back after a forced or planned outage, close monitoring and checking of CEP discharge parameters and Hotwell Specific conductivity should be done to rule out any possibility of condenser tube leakage during the shutdown period.	Operation/ Chemistry

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8.0 ACTIONS TO BE TAKEN IN CASE OF CYCLE CHEMISTRY DISTURBANCE DUE TO CTL OR INGRESS OF OTHER CONTAMINANTS.

8.1 ACTIVITIES TO BE CARRIED OUT BEFORE SHUTDOWN OF UNIT

Sl No	Activity	Responsibility
1.	i)In case the condensate CC/ DCC exceeds the normal value (Action Level 1) and other parameters (as defined 5.1 and 5.2) also begin to deviate, the Station Chemist will advise the unit control room to open the CBD to maintain boiler water chloride and silica levels within the recommended limits. ii)If the analytical values as mentioned in 5.2 show an increasing trend from their respective recommended values, condenser tube seepage/ leakage is confirmed. iii)Based on online and/or offline analysis, the site Chemistry shall confirm condenser tube leakage/seepage, or cycle chemistry disturbance and the same shall be recorded in the unit shift logbook. iv)In case of DM makeup water contamination, the service-in DM water storage tank or CST should be isolated immediately, and the standby DMWST/CST should be taken into service. The water quality of all CSTs must be checked, and if found contaminated, it should not be used for other units. Compliance with DM plant operation guidelines, as outlined in COS-OGN-CHEM-015, must be adhered to avoid such deviations. In case of a malfunction of a condensate polisher, it should be isolated immediately, and a standby CPU should be taken into service (if available). If a standby CPU is not available and the contamination level is high, the unit load must be reduced to ensure 100% condensate water flow through the available CPU. v)Refer to COS-OD-CHEM-001 for normal parameters limits.	Chemistry/ Operation
2.	If the boiler pH shows a downward trend, increase HP dosing to maintain the boiler pH within the range of 9.4 to 9.6. If required, diluted sodium hydroxide can be dosed along with TSP to maintain the boiler water pH in limit (Refer COS-OD-CHEM-001 for details). Overdosing of TSP/NaOH should be avoided. Increase ammonia dosing rate and maintain feed water pH 9.4-9.6(as all are ferrous units).	Chemistry/ Operation

3.	Both CPUs Should be kept in service to ensure 100% condensate flow through the CPUs and maintain the feed water and boiler water parameters to the maximum extent possible. The unit load should be reduced to ensure 100% condensate flow through the CPU in the following cases: i If the CPUs can't handle the total condensate flow. ii If a CPU service vessel is exhausted and resin transfer needs to be done. iii If standby CPU service vessel is not available.	Chemistry
4.	After detecting the condenser tube leakage or similar contamination, immediately isolate the unit from the common systems of the station, such as auxiliary PRDS and CST, to prevent contaminated steam or condensate from entering the other units.	Operation
5.	The recommended steam quality should be maintained by reducing drum pressure (if required) and operating CBD as advised by the site chemist. Refer to COS-OD-CHEM-001 for normal parameters limits.	Operation/ Chemistry
6.	Reduce superheater and reheater spray to minimum or stop in the affected unit.	Operation
7.	i)Using specific conductivity trends from the right and left sections of the hotwell, identify the pass where leakage is suspected, and take immediate action to isolate the affected section by closing CW inlet and outlet valves and drain the condenser water-box completely. ii)Under any circumstances the leaking condenser pass should be isolated before Cation conductivity in CEP sample reaches 1.0 µS/cm. iii)After isolating the condenser pass, stop fresh makeup and condensate dumping. Observe changes in condensate water chemistry parameters- CC, Sodium, silica, and hardness. iv)Carry out candle test or helium leak detection test or acoustic test to identify and plug leaking tube(s). v)If the leaking tube cannot be identified and cooling water ingress continues, shut down the unit and perform a flood test after completing condenser jacking as detailed in Annexure-III.	Operation/ Chemistry/ TMD
8.	The LPT exhaust spray line connected to the condensate header (before CPU) should be stopped (if feasible) to prevent exposure to contaminated condensate water.	Operation

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9.	i)If the boiler pH, measured at 25°C, falls below 8.0 with a further descending trend, immediately proceed with the shutdown of the unit as per 6.0. ii)If the boiler pH, measured at 25°C, greater than 11.0 with a further increasing trend, immediately proceed with the shutdown of the unit as per 6.0. iii)When Main steam CC >1.0μS/cm (increasing trend) or sodium >20ppb, immediately proceed with the shutdown of the unit as per 6.0.	Operation/ Chemistry

8.2 ACTIVITIES TO BE CARRIED OUT IMMEDIATELY AFTER UNIT SHUTDOWN

SI No	Activity	Responsibility
1	Feed flow to boiler should be stopped to protect boiler from contamination. Hotwell to deaerator flow also to be stopped.	Operation
2	Keep only one CPU in service, and both manual and pneumatic bypass valves should be fully closed.	Operation/ Chemistry
3	Hotwell flushing should be initiated and continued until all contaminated water is completely removed.	Operation/ Chemistry
4	Common systems like emergency make up, normal make up & PRDS should be isolated immediately. The water quality of all DMWST and CSTs should be checked. If found contaminated, they must be drained and flushed with fresh DM water. Completion criteria: Chloride < 10 ppb, Sodium < 10 ppb, and Silica < 20 ppb.	Operation/ Chemistry
5	The LPT exhaust spray line connected to the condensate header (before CPU) should be stopped to prevent exposure to contaminated condensate water.	Operation
6	Boiler water pH should be raised to 10.0 with phosphate dosing or alkali (NaOH) dosing and soak for one hour with boiler recirculation pumps in service. Boiler water should be drained after natural cooling. Boiler should be filled with fresh ammoniated DM water directly through boiler fill pump and drained. Filling and draining with ammoniated DM water, pH>9.5 should be continued till chloride level below 200 ppb is achieved in the boiler water sample.	Operation/ Chemistry
7	After depressurizing the deaerator, fill it with ammoniated DM water (pH > 9.5) using the boiler fill pump. Circulate the water using the MDFBP for 15–30 minutes, then drain it completely. Ensure that the drain valve of the BFP suction header is fully open.	Operation/ Chemistry

TITLE	MANAGEMENT OF CONDENSER TUBE LEAKAGE IN SEA WATER COOLED CONDENSER
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	Filling and draining should be continued until the chloride level in the drain water reaches 50 ppb. In case of severe contamination due to condenser tube leakage or similar issues, a chloride level of 100–120 ppb may be considered acceptable for initiating the earliest flushing of the feedwater and boiler circuits.	
8	Establish the FST normal level using the boiler fill pump and start flushing the feed water heaters with ammoniated DM water (pH > 9.5), draining through the boiler bottom ring header drains/ LP drains. FST heating may be carried out during this process. Keep the drains of all heaters full open throughout the activity. This flushing should be performed for a minimum of one hour at the maximum possible flushing rate. After that, the boiler should be filled up to the normal drum level and circulated through the BCP for a minimum of 30 minutes before draining through its drain valves. The boiler filling and draining is to be repeated till boiler water drain sample chloride value achieved <200ppb.	Operation/ Chemistry
9	Kill the vacuum, isolate the seal steam, and stop the barring gear to perform hotwell draining and subsequent flushing to be done after criteria of barring stopping is achieved. If Condensate cation conductivity is very high, barring stopping and draining of hotwell draining to be done in consultation with the site TMD and Chemistry. CC-OS to be consulted.	Operation/ TMD/ Chemistry
10	i)Since the CW system cannot be completely isolated immediately after the unit shutdown due to the need to keep the CEP in service, all contaminated water in the hotwell must be completely drained after the criteria of barring stopping is achieved. ii)The CEP discharge header and contaminated circuits must be completely drained after draining of hotwell. iii)Drain the shell and tube sides of the LP and HP heaters to the maximum extent. iv)Drain the HPFT, LPFT, SDFT, Flash Box-1, Flash Box-2 loops, and similar systems by opening the drain lines after criteria of hotwell draining is achieved. v)Any other specific circuit should also be drained to remove the contaminated water.	Operation
11	Cleaning and calibration of hotwell conductivity meters should also be carried out during the shutdown, if required.	C&I/ Chemistry
12	LP and HP chemical metering tanks should be cleaned if contaminated water has been used as makeup.	Chemistry

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13	Flushing of steam and water sample lines, coolers, etc. (both local and SWAS) should be carried out after flushing the respective systems.	C&I/ Operation
14	The cation resin in all CC columns of SWAS and CPU service vessels should be replaced with regenerated cation resin. This activity should be completed before the start of the unit startup process.	C&I/ Chemistry
15	The CPU service vessels exposed to contaminated water must be regenerated using double the quantity of acid and caustic. Refer to COS-OGN-CHEM-020 for detailed guidelines. This activity should be completed before the start of the unit startup process. One fresh resin lot should also be kept ready for changeover.	Chemistry

8.3ACTIVITIES TO BE CARRIED OUT AFTER RECTIFICATION OF CTL/ ELIMINATION OF CONTAMINANT SOURCE

<table>
<tr><td>S.No</td><td>Activity</td><td>Responsibility</td></tr>
<tr><td>1.</td><td>Flushing of all CEPs and their discharge headers should be carried out using DM water through the condensate dumping line, while keeping the CEPs in service. This activity should be performed after vacuum pulling and once the turbine is placed on barring gear.
Completion criteria:
<table>
<tr><td colspan="2">Sample:Condensate water(Condenser should be under vacuum)</td></tr>
<tr><td>Parameter</td><td>Limit</td></tr>
<tr><td>Chloride ppb, Max</td><td>20</td></tr>
<tr><td>Sodium ppb, Max</td><td>20</td></tr>
<tr><td>Silica ppb, Max</td><td>30</td></tr>
<tr><td>Turbidity, NTU,Max</td><td>5</td></tr>
<tr><td>CC/DCC us/cm,Max</td><td>1.0</td></tr>
</table>
Note: The Online Analyzer must be kept in service for the condensate water sample to ensure continuous monitoring.</td><td>Operation/ Chemistry</td></tr>
<tr><td>2</td><td>i)One CPU is to be put in service with the bypass fully closed (this must be strictly ensured). The second and CPU should not be taken into service.
ii)Ammonia dosing should be done at the CPU outlet/ CEP discharge.
iii)Start flushing the condensate header up to the last LP heaters while keeping all atmospheric drain valves open.
iv)The condensate flow should be swung during the flushing activity for effective cleanup.
A. HW-D/A OPEN FLUSHING</td><td>Operation/ Chemistry</td></tr>
</table>

<table>
<tr>
<td></td>
<td>Conduct Hotwell -deaerator open flushing with all BFP suction valves in open condition & drain to UFT or similar tank (as per design). The BPFs suction drain sample also to be analysed.
Completion criteria:
<table>
<tr><td colspan="2">Sample: FST Drain water (Feed water o/l)</td></tr>
<tr><td>Parameters</td><td>Limit</td></tr>
<tr><td>Chloride, ppb,Max</td><td>20</td></tr>
<tr><td>Sodium, ppb,Max</td><td>10</td></tr>
<tr><td>Silica, ppb,Max</td><td>30</td></tr>
<tr><td>Turbidity, NTU,Max</td><td>5</td></tr>
<tr><td>p H at 25 deg C</td><td>8.8-9.3/9.2-9.6/9.5-9.7
Refer to COS-OD-Chem-001</td></tr>
</table>
HW-D/A CLOSED CIRCULATION
Start HW-D/A closed circulation and continue until the recommended parameters, as outlined in COS-OGN-SYST-065, are achieved.
The chloride levels also to be monitored in the FST water sample and should be <10 ppb.
BFPs should be operated to flush their suction, discharge, and recirculation headers.</td>
<td></td>
</tr>
<tr>
<td>3</td>
<td>Start flushing the feed water heaters with FST water, regardless of the deaerator temperature, and drain through the boiler bottom ring header drains/ LP drains. FST heating and pegging should also be done simultaneously. Keep the drains of all heaters open throughout the activity.
This flushing should be carried out for at least one hour at the maximum possible flushing/draining rate.</td>
<td>Operation/ Chemistry</td>
</tr>
<tr>
<td>4</td>
<td>The boiler filling, circulation and draining is to be repeated till boiler water drain sample chloride value achieved <200ppb.</td>
<td>Operation/ Chemistry</td>
</tr>
<tr>
<td>5</td>
<td>i)During light up, DM make up should be taken into deaerator using boiler fill pump with CEP to deaerator line in isolated condition by closing deaerator control valve. One CPU should be kept in service with the bypass valve fully closed.
ii)The boiler hot cleanup is to be done as per COS-OGN-SYST-065 guidelines. HP dosing should be done and maintain boiler water pH 9.4-9.6.
iii)The boiler should be completely hot drained at 10-15 kg/cm2 drum pressure after BLU.
iv)After BLU, the boiler water chemistry is to be maintained in accordance with COS-OD-CHEM-001.
v)Turbine rolling clearance should be accorded only after achieving MS chloride < 5ppb (as per Ion Chromatograph results) along with other recommended parameters outlined in COS-OGN-SYST-065.
vi)100% Condensate flow through CPU is to be ensured.</td>
<td>Operation/ Chemistry</td>
</tr>
</table>

	vii)Gradually reduce phosphate and chloride level by boiler blow down ensuring that boiler pH is maintained to facilitate increase in drum pressure.	
6	Steam and water sample analysis should be carried out every two hours for all recommended chemistry parameters.	Chemistry
7	All chemical parameters and operational activities must be recorded from the onset of the condenser tube leakage/disturbance in cycle chemistry parameters until normalization of the parameters. These records should be maintained for future reference. Documentation related to the condenser tube leakage/Chemistry parameters disturbance shall be maintained as per the format provided in **Annexure-I**.	Operation/ Chemistry
8	The chemistry parameters for each cleanup step should be recorded in the standard templates as specified in **COS-OGN-SYST-065**.	Chemistry

9.0 RECORDS TO BE MAINTAINED

Sl No	**Record Description**	**Maintained by**	**Frequency**	**Remarks**
1	Reports as per Annexure-I &II	Chemistry	For each event	
2	Reports for unit startup cleanup parameters	Chemistry	For each event	
3	Excursion durations for each parameter during unit operation at different Action Levels, as per 6.1	Chemistry	For each event	
4	Details of condenser tubes plugged.	TMD	For each event	
5	Status of Sec. 7.0 recommendations	Operation	Once/year	

10.0 DISTRIBUTION OF LMI

A soft copy of the LMI shall be uploaded in the EEMG dept. website on Simhadri intranet LAN by the EEMG dept. The uploaded soft copy of the document is considered as Controlled Copy.

11.0 REVIEW OF LMI

Head of Chemistry is responsible for put up approval from Competent Authority and for revision of LMI on a 2-yearly basis or as and when required.

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Check List-I

FORMAT FOR CTL/ CYCLE CHEMISTRY DISTURBANCE

Station: …………………… Unit:………….. Date:…………..

Date and Time of Cycle Chemistry Disturbance Observation:……………………

Sl. No	DETAILS	STATUS& DATE (ACTIVITY START &End Time)
1	Unit Load / Drum or Separator Pressure/ Condensate Flow at the Time of CTL / Cycle Chemistry Disturbance.	
2	Type of Cooling Water (Fresh/Sea water)	
3	Condenser tube metallurgy	
4	Whether any permanent CW treatment program is in place	
5	Which chemical parameter(s) showed deviation, confirming the occurrence of condenser tube leakage in the condensate system?	
6	Availability of online instruments for core parameters	
7	Excursion durations for each parameter during unit operation at different Action Levels, as per Clause 6.0-C	
8	Status of CPU Bypass valve (For Units equipped with CPU)	
9	Immediate action taken: i)Operational Steps ii)Action taken to maintain chemical parameters	
10	System affected by CTL / Contaminant Ingress: Condensate System / Feed Water System / Boiler / Steam Circuit	
11	Status of Isolation of Common Systems (PRDS, CST, etc.)	
12	Is this the first instance of tube leakage? (Past history details, if applicable, should be mentioned.)	
13	Was the unit taken under shutdown?	
14	Was pass isolation performed? If yes, please provide details, along with information on subsequent activities such as flood tests or dye tests conducted.	

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15	No of condenser tubes plugged	
16	Details of cleanup activities	
17	Restoration of normal operating parameters of unit.	
Any other remarks:		

Signature: (Chemistry) (Operation) (TMD)

NOTE:

1. In addition, chronological details of the chemical parameters must be recorded during condenser tube leakage or cycle chemistry disturbance, as per the prescribed format in Annexure-II

2. The filled template should be uploaded and maintained on the site intranet, with a copy sent to CC-OS Chemistry.

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Check List-II

FORMAT FOR RECORDING CYCLE CHEMISTRY PARAMETERS DURING CTL/ INGRESS OF CONTAMINANTS

Station: Unit: Load: Drum/Separator Pressure:

Date & Time:

Sample Name	**Parameters**	**Limiting Value***	**Actual Value****	**Remarks**
HOT WELL (L/R)	Sp Cond(μS/cm)			
CEP	ACC (μS/cm)/DCC			
	Na (ppb)			
	Sp Cond(μS/cm)			
	Chloride (ppb)			
	DO (ppb)			
	pH			
	Silica (ppb)			
	Hardness(ppb)			
CPU (O/L)-VESSEL WISE	ACC (μS/cm)			
	Na (ppb)			
	Chloride (ppb)			
	Silica (ppb)			
	pH			
	Sp Cond(μS/cm)			
FEED WATER	ACC (μS/cm)			
	Chloride (ppb)			
	Na (ppb)			
	Silica(ppb)			
	DO(ppb)			

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	pH			
	Sp Cond(μS/cm)			
BOILER DRUM	ACC (μS/cm)			
	Sp Cond (μS/cm)			
	Chloride (ppb)			
	Silica (ppb)			
	Phosphate (ppm)			
	pH			
MAIN STEAM	ACC (μS/cm)			
	Na (ppb)			
	Chloride (ppb)			
	Silica (ppb)			
	pH			
	Sp Cond (μS/cm)			
HRH	pH			
	Sp Cond (μS/cm)			
	ACC (μS/cm)			
	Silica (ppb)			
	Sodium (ppb)			
	Chloride (ppb)			

Refer **COS-OD-CHEM-001** for parameters limit.

** Wherever both online and offline values are available for parameters, both offline and online values are to be recorded as long as the unit is on bar.

Signature: **(Chemistry)**

TITLE	MANAGEMENT OF CONDENSER TUBE LEAKAGE IN SEA WATER COOLED CONDENSER
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ANNEXURE-I

CONDENSER FLOOD TEST PROCEDURE

S.No	ACTIVITY	RESPONSIBILITY
1	Stopping of TG barring gear at HP casing temperature of less than200 Deg C with lube oil & jacking oil system in service with intermittent manual barring.	HOD (Operation).
2	Stopping of both running CW Pumps. Closing of Condenser CW inlet& outlet BFVs & draining of condenser water boxes.	HOD (Operation).
3	Jacking of condenser bottom springs / providing additional support in hot well bottom plate as per OEM guidelines.	HOD (TMD).
4	Closing of all MAL drains/Extractions drains/MS/HRH/CRH line drains etc. connected to condenser flash box/flask tank.	HOD (Operation).
5	Filling of DM water in hot well steam space up to one meter above tube bundle.	HOD (Operation).
6	Opening of condenser water box manhole covers after obtaining PTW. Identification of leaking condenser tubes & plugging. If required air drying of condenser tubes to identify leakage tube.	HOD (TMD).

NOTE: Any deviation to the above procedure clearance may be taken from COS-Steam Turbine group.